

Data Session 9 Key, part 1: A Poll That Is Precise and Wrong

84-355 Democracy's Data | Session 9 — Polls, models, and markets | Instructor copy — do not distribute

Jonathan Cervas

August 11, 2026

All Part 9 code is given to students. Grade the interpretation.

Every number below was run. **Note that this lab is stochastic** — the student copy sets `set.seed(2026)`, so their numbers will match each other, but if a student re-runs a chunk interactively they will get slightly different values. That is not an error, and it is worth saying so in class before somebody panics.

The lab in one sentence

A margin of error describes sampling and nothing else — so a poll can be made arbitrarily precise and remain arbitrarily wrong, which is the actual condition of American polling and the argument of the Bailey reading.

The demonstration

```
answers <- ifelse(population == "Harris", 1.20, 1.00)
hl <- data.frame()
for (k in c(1000, 10000, 100000)) {
  s <- sample(population, k, prob = answers, replace = TRUE)
  est <- 100 * mean(s == "Harris"); m <- 1.96 * sqrt(0.25/k) * 100
  hl <- rbind(hl, data.frame(n = k, est = est, moe = m,
                             err = est - 100*truth,
                             mult = abs(est - 100*truth) / m))
  cat(sprintf("n = %6d  estimate %.2f  reports +/-%.2f  wrong by %+.2f  (%.0fx its own MoE)\n",
              k, est, m, est - 100*truth, abs(est - 100*truth)/m))
}
```

```
#> n =   1000  estimate 53.20  reports +/-3.10  wrong by +3.95  (1x its own MoE)
#> n =  10000  estimate 53.88  reports +/-0.98  wrong by +4.63  (5x its own MoE)
#> n = 100000  estimate 53.59  reports +/-0.31  wrong by +4.34  (14x its own MoE)
```

At $n = 100,000$ the poll reports ± 0.31 and misses by **4.34 points** — roughly **14 times** its stated precision. The asymptotic value is **53.8%**, not 49.25: it converges on the wrong answer.

```
round(100 * (truth * 1.20) / (truth * 1.20 + (1 - truth)), 2)
```

```
#> [1] 53.8
```

This is the whole lab. If a student leaves with one sentence, it should be that more data fixes variance and does nothing to bias.

Option A — How big does the bias have to be?

Verified output

```
bA <- data.frame()
for (g in c(0.05, 0.10, 0.20, 0.40)) {
  a <- ifelse(population == "Harris", 1 + g, 1)
  e <- replicate(300, 100*mean(sample(population, 1000, prob = a, replace = TRUE) == "Harris"))
  asym <- 100 * (truth*(1+g)) / (truth*(1+g) + (1-truth))
  bA <- rbind(bA, data.frame(gap = 100*g, bias = mean(e) - 100*truth,
                             asym = asym - 100*truth))
  cat(sprintf(" gap %2.0f%%: bias %+.2f (asymptotic %+.2f)\n",
              100*g, mean(e) - 100*truth, asym - 100*truth))
}
```

```
#> gap 5%: bias +1.03 (asymptotic +1.22)
#> gap 10%: bias +2.45 (asymptotic +2.38)
#> gap 20%: bias +4.63 (asymptotic +4.55)
#> gap 40%: bias +8.42 (asymptotic +8.35)
```

response gap	bias	vs. the real 1.5-pt margin
5%	+1.03	comparable to the entire margin
10%	+2.45	larger than the margin
20%	+4.63	three times the margin
40%	+8.42	absurd, and invisible in the MoE

What a good answer says

3/4: Reports that even a 5% gap produces about a point of bias, and notices that the advertised ± 3.1 never changes.

4/4: Puts it against the real election. **2024 was decided by 1.5 points.** A 5% difference in willingness to answer the phone — which no pollster could detect and which sounds like nothing — is of the same order as the entire result. So the honest reading of a poll showing a 1.5-point race is not “candidate A is narrowly ahead” but “this instrument cannot resolve this question.”

4/4, best: notices that the bias is roughly **linear in the gap** while the margin of error is **not affected at all**, so the two quantities are not commensurable and reporting only one of them is a choice. Asks what a poll would have to publish for a reader to judge it — and lands on something like “our assumed response model,” which nobody publishes.

Common stumble: treating the 5% row as reassuring because $1.03 < 3.1$. The comparison that matters is 1.03 against the *margin of the election* (1.5 points), not against the MoE.

Option B — Does averaging help?

Verified output

```
a <- ifelse(population == "Harris", 1.20, 1)
avg <- data.frame()
for (np in c(5, 20, 200)) {
  h <- replicate(np, 100*mean(sample(population, 1000) == "Harris"))
  s <- replicate(np, 100*mean(sample(population, 1000, prob = a, replace = TRUE) == "Harris"))
  avg <- rbind(avg, data.frame(np = np, honest = mean(h), skewed = mean(s)))
  cat(sprintf(" %3d polls: honest avg %.2f | skewed avg %.2f\n", np, mean(h), mean(s)))
}
```

```
#>    5 polls: honest avg 49.72 | skewed avg 54.04
#>   20 polls: honest avg 49.84 | skewed avg 53.87
#>  200 polls: honest avg 49.21 | skewed avg 53.89
```

Averaging honest polls converges on **49.21** — the truth is 49.25. Averaging skewed polls sits at **53.87–54.04** and stays there however many you add.

What a good answer says

3/4: Reports both averages and states that averaging fixes noise but not bias.

4/4: Answers the aggregator question properly. A polling average’s tight interval measures **the agreement among pollsters**, not the distance from the truth. Those are the same thing only if the polls’ errors are independent — and they are not, because every pollster is drawing from the same shrinking pool of people who answer surveys and correcting with broadly similar methods. Agreement is evidence of shared method, not of accuracy.

4/4, best: notices this predicts a specific failure mode — that when polls miss they should all miss *the same way at once*, which is exactly the observed pattern in 2016 and 2020, and why state-level errors are correlated rather than cancelling.

Option C — Weighting on the wrong thing

Verified output

```
N2 <- 200000
harris <- rbinom(N2, 1, truth)
trust <- rbinom(N2, 1, ifelse(harris == 1, 0.75, 0.45))
tv <- 100 * mean(harris)
wt <- data.frame()
for (st in c(0.1, 0.35, 0.8)) {
  resp <- ifelse(trust == 1, 1, st)
  j <- sample(N2, 5000, prob = resp, replace = TRUE)
  ss <- data.frame(h = harris[j], t = trust[j]); tgt <- mean(trust)
  w <- ifelse(ss$t == 1, tgt/mean(ss$t), (1-tgt)/(1-mean(ss$t)))
  wt <- rbind(wt, data.frame(st = st, raw = 100*mean(ss$h) - tv,
                             wtd = 100*weighted.mean(ss$h, w) - tv))
  cat(sprintf(" strength %.2f: raw %+.2f error | weighted %+.2f error\n",
              st, 100*mean(ss$h) - tv, 100*weighted.mean(ss$h, w) - tv))
}

#> strength 0.10: raw +10.01 error | weighted -0.19 error
#> strength 0.35: raw +6.63 error | weighted +0.41 error
#> strength 0.80: raw +1.07 error | weighted -0.86 error
```

non-response strength	raw error	after weighting
0.10 (severe)	+10.01	-0.19
0.35	+6.63	+0.41
0.80 (mild)	+1.07	-0.86

What a good answer says

3/4: Reports that weighting rescues the estimate at every strength — every row lands within 0.9 of the truth — and that the raw error grows as non-response worsens.

4/4: States the condition plainly — **weighting works because we can see trust**. The variable is in the dataset. If the thing driving non-response were something the pollster does not measure, the correction is unavailable and the raw error stands. Weighting is not a general remedy for non-response; it is a remedy for *known* non-response.

4/4, best: connects to the industry’s actual history. After 2016 pollsters discovered they had not been weighting by education and added it. In 2020 they missed again, in the same direction, because the next unmeasured variable was already waiting. The problem is not that pollsters are careless; it is that the variable you have not thought of is invisible by construction.

Watch for: students concluding “so weighting is useless.” It took +10.01 to -0.19. The point is narrower and more interesting than that.

Option D — Their own question

The best version is the trick question in the prompt: how large would a poll have to be for its margin of error to fall below the bias? At a 20% gap the bias is 4.63 points, and MoE drops below that at roughly $n = 448$ — so **the answer is “almost immediately, and it does not help.”** Any poll worth running is already in the regime where bias dominates.

Grading

Four-point scale from the syllabus.

- **Never penalise code.** It is your code, and this one is stochastic — do not penalise numbers that differ from the key in the second decimal.
- **The move that earns a 4** is comparing bias to the *election margin* rather than to the margin of error. Students who stay inside the MoE frame have reproduced the error the lab is about.
- **Reward caution about Tuesday.** A student who says “I no longer know what to make of the polls I am reading” has understood the lab, and should be told that this is the correct state to be in, not a failure to reach a conclusion.

Anticipated questions

- “*So are polls useless?*” — No. They are excellent at large differences and poor at small ones, and 2024 was a small one. A poll showing a 20-point race is almost certainly right about who is ahead.
- “*Why not just poll more people?*” — Part 6. This is the question the lab exists to answer.
- “*Is the 20% gap realistic?*” — It is invented, and the lab says so. Real response rates are under 1% and the composition of that 1% is not measured directly. The shape transfers; the size does not.
- “*Why is this a simulation when every other lab uses real data?*” — Because you cannot see a poll fail without knowing the truth. Worth saying out loud; several students will ask.
- “*Did the 2024 polls actually miss?*” — Nationally they were closer than 2016 or 2020, and still understated Trump in most states. Do not overstate this in either direction; the honest version is that the direction of error has been consistent for three cycles and nobody has a settled explanation.

Connections

- **Session 5 (media attention)** — attention and response are both behaviours we can measure; belief is not. Same limit, different instrument.
- **Session 10, next Thursday** — students will have the answers. The forecast they read is now testable, and Option B’s point about correlated errors is what to watch for in the state-level misses.
- **Session 3 (§203) and Session 2 (apportionment)** — both apply a sharp rule to an estimated quantity. This lab is the same structure with the estimate’s error made visible.